



Wetford Co. 4-2-15
Cadillac WWTP JHW

Cadillac Utilities Department
200 Lake St.
Cadillac, MI 49601
231-775-0181
231-775-8755

WATER RESOURCES
DIVISION

MAR 31 2015

CADILLAC DISTRICT

March 31, 2015

Mr. Jamie Wade, P.E.
Michigan Department of Environmental Quality
Cadillac District Office
120 West Chapin Street
Cadillac, Michigan 49601

Re: 2014'S POLLUTANT MINIMIZATION PLAN YEARLY STATUS REPORT

Dear Mr. Wade,

Enclosed, is the Cadillac WWTP 2014 Pollutant Minimization Plan Yearly Status Report. I have enclosed the Influent and Effluent Mercury results for 2014, also included is a report of mercury levels in Cadillac's Biosolids.

There was no industry that came to town that had a potential of new mercury source and all of the previous potential mercury sources have not changed their processes. Because of this there is no updated list of potential mercury sources for the City of Cadillac.

A close Examination of the influent and effluent data shows "less than detect" values for mercury concentration for 2014. This level of mercury meets the goal of the Pollutant Minimization Program. We will continue to monitor the level of mercury in our effluent per our NPDES permit.

Please feel free to call if you have any questions.

Sincerely,

A handwritten signature in black ink, appearing to read "Hamid Rahbarnoohi", with a long horizontal flourish extending to the right.

Hamid Rahbarnoohi, Water Resources Supervisor
Cadillac WWTP
Phone #: (231) 775- 2841
Fax #: (231) 775- 2876
E-mail: hrahbarnoohi@cadillac-mi.net
Cc: File -Mercury Minimization reports

A handwritten mark or signature in the bottom right corner of the page, consisting of a few vertical and diagonal strokes.

List of Potential Mercury Sources 2014

1. AAR Factory	AAR 201 Haynes Street
2. AAR Rinse Tank	AAR 201 Haynes Street
3. AAR 6 th	AAR 6 th Street
4. Avon Automotive	Avon 603 West 7 th Street
5. BorgWarner	BorgWarner 1100 Wright Street
6. Hutchinson^	600 Seventh Street
7. Michigan Rubber	Michigan Rubber 1200 Eighth Street
8. Rec Boat Holdings*	Rec Boat Holdings 609 West 13 th Street
9. Rec Boat Holding Sport*	Rec Boat Holdings Sport 925 Frisbie Street
10. Rec Boat Holdings Trailer*	Rec Boat Holdings Trailer 1500 Ron Wilson
11. Cadillac Castings Inc.	CCI 1500 Fourth Street
12. Wexford County Landfill, LLC**	900 N. US HWY 131, Manton, MI 49663
13. Cadillac Renewable Energy	CRE, 1525 Frank Miltner
14. Mercy Hospital	Mercy Hospital 400 Hobart Street
15. FIAMM	FIAMM Technologies 946 Frisbie Street
16. ARVCO	ARVCO 1355 Marty Paul Street
17. Piranha Hose	Piranha Hose 2500 Raymond A. Wiegler St.
18. Avon Protection	Avon Protection 503 8 th Street
19. B&P Manufacturing	8051 East 34 Road, Cadillac, MI 49601
20. Spencer Plastics	2300 Gary E. Schwach St, Cadillac, MI

^Hutchison was formally known as Paulstra

*Rec Boat Holdings formerly known as 4-Winns

**Wexford County Landfill, LLC is owned currently by American Waste, formerly owned by Wexford County

City of Cadillac

Mercury, Influent and Effluent concentration through Cadillac Wastewater Treatment Plant (2014)

Date	Effluent, Hg, ng/L, EPA Method 1631	Influent Hg, mg/L, Low level Hg
11/12/14	<0.5	<0.0002
8/6/14	<0.5	<0.0002
5/6/14	<0.5	<0.0002
2/3/14	<0.5	<0.0002

Mercury Concentration in Biosolids, 2014

Sample Date	Mercury Conc. mg of Hg/Kg total solids
11/03/14	0.89
7/10/14	0.48
3/31/14	0.72

LABORATORY DETAIL REPORT

Client: *City of Cadillac, Water Resources*

KAR Project No. : **408941**

Date Reported : **11/25/14**

Attest: 
David R. Alkema, Lab Manager

Project

Description: *Analysis of one aqueous sample for Mercury Minimization.*

Sample ID : **"Influent"**

Sampled By : *AV of City of Cadillac, Water Resources*

Date Received : **11/13/14**

Sample Date : **11/11/14**

Sample Type : **aqueous**

Sample Time : **0820**

KAR Sample No. : **408941-01**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
<i>Prep, Hg</i>	<i>Completed</i>		<i>EPA 245.2</i>	<i>11/21/14</i>	<i>BPK</i>	
<i>Mercury, total, low level</i>	<i><0.0002</i>	<i>mg/L</i>	<i>EPA 245.2</i>	<i>11/24/14</i>	<i>BPK</i>	

LABORATORY DETAIL REPORT

Client: *City of Cadillac, Water Resources*

KAR Project No. : **408942**

Date Reported : **11/25/14**

Attest:


David R. Alkema, Lab Manager

Project

Description: *Analysis of four aqueous samples for ultra-trace mercury.*

Sample ID : **"Effluent"**

Sampled By : *AV of City of Cadillac, Water Resources*

Date Received : **11/13/14**

Sample Date : **11/12/14**

Sample Type : **aqueous**

Sample Time : **1248**

KAR Sample No. : **408942-01**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Prep, 1631	Completed		EPA 1631E	11/19/14	MHK	
Mercury by EPA 1631	<0.5	ng/L	EPA 1631E	11/20/14	MHK	

Sample ID : **"Effluent, Field Duplicate"**

Sampled By : *AV of City of Cadillac, Water Resources*

Date Received : **11/13/14**

Sample Date : **11/12/14**

Sample Type : **aqueous**

Sample Time : **1250**

KAR Sample No. : **408942-02**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Prep, 1631	Completed		EPA 1631E	11/19/14	MHK	
Mercury by EPA 1631	<0.5	ng/L	EPA 1631E	11/20/14	MHK	

Sample ID : **"Field Blank"**

Sampled By :

Date Received : **11/13/14**

Sample Date : **11/12/14**

Sample Type : **aqueous**

Sample Time :

KAR Sample No. : **408942-03**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Prep, 1631	Completed		EPA 1631E	11/19/14	MHK	
Mercury by EPA 1631	<0.2	ng/L	EPA 1631E	11/20/14	MHK	Below quantitation limit; result is suspect.

Sample ID : **"Trip Blank"**

Sampled By :

Date Received : **11/13/14**

Sample Date :

Sample Type : **Trip Blk**

Sample Time :

KAR Sample No. : **408942-04**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Prep, 1631	Completed		EPA 1631E	11/19/14	MHK	
Mercury by EPA 1631	<0.2	ng/L	EPA 1631E	11/20/14	MHK	Below quantitation limit; result is suspect.

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LABORATORY DETAIL REPORT

Client: *City of Cadillac, Water Resources*

KAR Project No. : **408942**

Date Reported : **11/25/14**

Attest: 
David R. Alkema, Lab Manager

Project

Description: *Analysis of four aqueous samples for ultra-trace mercury.*

Sample ID : **Laboratory Method Blank**

Sampled By :

Date Received : **11/13/14**

Sample Date : **11/13/14**

Sample Type : **LMB**

Sample Time :

KAR Sample No. : **408942-05**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Prep, 1631	Completed		EPA 1631E	11/19/14	MHK	
Mercury by EPA 1631	<0.2	ng/L	EPA 1631E	11/20/14	MHK	Below quantitation limit; result is suspect.

LABORATORY DETAIL REPORT

Client: *City of Cadillac, Water Resources*

KAR Project No. : **405714**

Date Reported : **08/15/14**

Attest: 
David R. Alkema, Lab Manager

Project

Description: *Analysis of one aqueous sample for Mercury Minimization.*

Sample ID : **"Influent"**

Sampled By : *AJV of City of Cadillac, Water Resources*

Date Received : **08/07/14**

Sample Date : **08/06/14**

Sample Type : **aqueous**

Sample Time : **0758**

KAR Sample No. : **405714-01**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
<i>Prep. Hg</i>	<i>Completed</i>		<i>EPA 245.2</i>	<i>08/11/14</i>	<i>ALK</i>	
<i>Mercury, total, low level</i>	<i><0.0002</i>	<i>mg/L</i>	<i>EPA 245.2</i>	<i>08/11/14</i>	<i>ALK</i>	

LABORATORY DETAIL REPORT

Client: *City of Cadillac, Water Resources*

KAR Project No. : **405716**

Date Reported : **08/21/14**

Attest: 
David R. Alkema, Lab Manager

Project

Description: *Analysis of four aqueous samples for ultra-trace mercury.*

Sample ID : **"Effluent"**

Sampled By : *AJV of City of Cadillac, Water Resources*

Date Received : **08/07/14**

Sample Date : **08/06/14**

Sample Type : **aqueous**

Sample Time : **1013**

KAR Sample No. : **405716-01**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Prep, 1631	Completed		EPA 1631E	08/19/14	JHB	
Mercury by EPA 1631	<0.5	ng/L	EPA 1631E	08/20/14	JHB	

Sample ID : **"Effluent, Field Duplicate"**

Sampled By : *AJV of City of Cadillac, Water Resources*

Date Received : **08/07/14**

Sample Date : **08/06/14**

Sample Type : **aqueous**

Sample Time : **1014**

KAR Sample No. : **405716-02**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Prep, 1631	Completed		EPA 1631E	08/19/14	JHB	
Mercury by EPA 1631	<0.5	ng/L	EPA 1631E	08/20/14	JHB	

Sample ID : **"Field Blank"**

Sampled By : *AJV of City of Cadillac, Water Resources*

Date Received : **08/07/14**

Sample Date : **08/06/14**

Sample Type : **aqueous**

Sample Time :

KAR Sample No. : **405716-03**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Prep, 1631	Completed		EPA 1631E	08/19/14	JHB	
Mercury by EPA 1631	<0.5	ng/L	EPA 1631E	08/20/14	JHB	

Sample ID : **"Trip Blank"**

Sampled By :

Date Received : **08/07/14**

Sample Date :

Sample Type : **Trip Blk**

Sample Time :

KAR Sample No. : **405716-04**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Prep, 1631	Completed		EPA 1631E	08/19/14	JHB	
Mercury by EPA 1631	<0.5	ng/L	EPA 1631E	08/20/14	JHB	

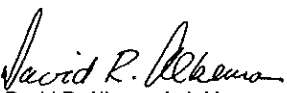
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LABORATORY DETAIL REPORT

Client: *City of Cadillac, Water Resources*

KAR Project No. : **403027**

Date Reported : **05/20/14**

Attest: 
David R. Alkema, Lab Manager

Project

Description: *Analysis of one aqueous sample for Mercury Minimization.*

Sample ID : **"Influent"**

Sampled By : *AV of City of Cadillac, Water Resources*

Date Received : **05/07/14**

Sample Date : **05/06/14**

Sample Type : **aqueous**

Sample Time : **1235**

KAR Sample No. : **403027-01**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
<i>Prep, Hg</i>	<i>Completed</i>		<i>EPA 245.2</i>	<i>05/12/14</i>	<i>BPK</i>	
<i>Mercury, total, low level</i>	<i><0.0002</i>	<i>mg/L</i>	<i>EPA 245.2</i>	<i>05/13/14</i>	<i>BPK</i>	

LABORATORY DETAIL REPORT

Client: *City of Cadillac, Water Resources*

KAR Project No. : **403025**

Date Reported : **05/20/14**

Attest: 
David R. Alkema, Lab Manager

Project

Description: *Analysis of four aqueous samples for ultra-trace mercury.*

Sample ID : **"Effluent"**

Sampled By : *AV of City of Cadillac, Water Resources*

Date Received : **05/07/14**

Sample Date : **05/06/14**

Sample Type : **aqueous**

Sample Time : **1036**

KAR Sample No. : **403025-01**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Prep, 1631	Completed		EPA 1631E	05/13/14	ZZW	
Mercury by EPA 1631	<0.5	ng/L	EPA 1631E	05/14/14	ZZW	

Sample ID : **"Effluent, Field Duplicate"**

Sampled By : *AV of City of Cadillac, Water Resources*

Date Received : **05/07/14**

Sample Date : **05/06/14**

Sample Type : **aqueous**

Sample Time : **1036**

KAR Sample No. : **403025-02**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Prep, 1631	Completed		EPA 1631E	05/13/14	ZZW	
Mercury by EPA 1631	<0.5	ng/L	EPA 1631E	05/14/14	ZZW	

Sample ID : **"Field Blank"**

Sampled By : *AV of City of Cadillac, Water Resources*

Date Received : **05/07/14**

Sample Date : **05/06/14**

Sample Type : **aqueous**

Sample Time :

KAR Sample No. : **403025-03**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Prep, 1631	Completed		EPA 1631E	05/13/14	ZZW	
Mercury by EPA 1631	<0.5	ng/L	EPA 1631E	05/14/14	ZZW	

Sample ID : **"Trip Blank"**

Sampled By :

Date Received : **05/07/14**

Sample Date :

Sample Type : **Trip Blk**

Sample Time :

KAR Sample No. : **403025-04**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Prep, 1631	Completed		EPA 1631E	05/13/14	ZZW	
Mercury by EPA 1631	<0.5	ng/L	EPA 1631E	05/14/14	ZZW	

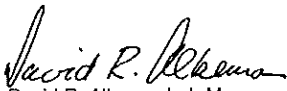
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LABORATORY DETAIL REPORT

Client: *City of Cadillac, Water Resources*

KAR Project No. : **400730**

Date Reported : **02/17/14**

Attest: 
David R. Alkema, Lab Manager

Project

Description: *Analysis of one aqueous sample for Mercury Minimization.*

Sample ID : **"Influent"**

Sampled By : *AJV of City of Cadillac, Water Resources*

Date Received : **02/04/14**

Sample Date : **02/03/14**

Sample Type : **aqueous**

Sample Time : **1022**

KAR Sample No. : **400730-01**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Prep. Hg	Completed		EPA 245.2	02/12/14	ALK	
Mercury, total, low level	<0.0002	mg/L	EPA 245.2	02/12/14	ALK	

LABORATORY DETAIL REPORT

Client: *City of Cadillac, Water Resources*

KAR Project No. : **400737**

Date Reported : **02/18/14**

Attest:


David R. Alkema, Lab Manager

Project

Description: *Analysis of four aqueous samples for ultra-trace mercury.*

Sample ID : **"Effluent"**

Sampled By : *AJV of City of Cadillac, Water Resources*

Date Received : **02/04/14**

Sample Date : **02/03/14**

Sample Type : **aqueous**

Sample Time : **1336**

KAR Sample No. : **400737-01**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Prep, 1631	Completed		EPA 1631E	02/10/14	ZZW	
Mercury by EPA 1631	<0.5	ng/L	EPA 1631E	02/11/14	ZZW	

Sample ID : **"Effluent, Field Duplicate"**

Sampled By : *AJV of City of Cadillac, Water Resources*

Date Received : **02/04/14**

Sample Date : **02/03/14**

Sample Type : **aqueous**

Sample Time : **1338**

KAR Sample No. : **400737-02**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Prep, 1631	Completed		EPA 1631E	02/10/14	ZZW	
Mercury by EPA 1631	<0.5	ng/L	EPA 1631E	02/11/14	ZZW	

Sample ID : **"Field Blank"**

Sampled By : *AJV of City of Cadillac, Water Resources*

Date Received : **02/04/14**

Sample Date : **02/03/14**

Sample Type : **aqueous**

Sample Time :

KAR Sample No. : **400737-03**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Prep, 1631	Completed		EPA 1631E	02/10/14	ZZW	
Mercury by EPA 1631	<0.5	ng/L	EPA 1631E	02/11/14	ZZW	

Sample ID : **"Trip Blank"**

Sampled By :

Date Received : **02/04/14**

Sample Date :

Sample Type : **Trip Blk**

Sample Time :

KAR Sample No. : **400737-04**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Prep, 1631	Completed		EPA 1631E	02/10/14	ZZW	
Mercury by EPA 1631	<0.5	ng/L	EPA 1631E	02/14/14	ZZW	

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LABORATORY DETAIL REPORT

Client: *City of Cadillac, Water Resources*

KAR Project No. : **408610**

Date Reported : **11/18/14**

Attest: 
David R. Alkema, Lab Manager

Project

Description: *Analysis of one sludge sample for Fall Biosolids.*

Sample ID : **"Digester #4"**

Sampled By : *KB of City of Cadillac, Water Resources*

Date Received : **11/04/14**

Sample Date : **11/03/14**

Sample Type : **sludge**

Sample Time : **1150**

KAR Sample No. : **408610-01**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Prep, Hg	Completed		EPA 7471A	11/13/14	BPK	
Prep, metals	Completed		EPA 3050	11/06/14	MHK	
Aluminum, total	3470	mg/kg total solids	EPA 6010B	11/14/14	JHB	Detection limit = 47
Arsenic, total	<19	mg/kg total solids	EPA 200.8	11/17/14	NHM	
Cadmium, total	<10	mg/kg total solids	EPA 200.8	11/17/14	NHM	
Calcium, total	21,100	mg/kg total solids	EPA 6010B	11/14/14	JHB	Detection limit = 47
Chromium, total	67.8	mg/kg total solids	EPA 6010B	11/14/14	JHB	Detection limit = 4.7
Copper, total	697	mg/kg total solids	EPA 6010B	11/14/14	JHB	Detection limit = 9.3
Lead, total, by ICP	28.1	mg/kg total solids	EPA 6010B	11/14/14	JHB	Detection limit = 23.2
Magnesium, total	3270	mg/kg total solids	EPA 6010B	11/14/14	JHB	Detection limit = 47
Mercury, total	0.89	mg/kg total solids	EPA 7471A	11/13/14	BPK	Detection limit = 0.11
Molybdenum, total	17.8	mg/kg total solids	EPA 6010B	11/14/14	JHB	Detection limit = 9.3
Nickel, total	24.1	mg/kg total solids	EPA 6010B	11/14/14	JHB	Detection limit = 9.3
Potassium, total	2500	mg/kg total solids	EPA 6010B	11/14/14	JHB	Detection limit = 47
Selenium, total	<10	mg/kg total solids	EPA 200.8	11/17/14	NHM	
Silver, total	2.7	mg/kg total solids	EPA 6010B	11/13/14	BPK	Detection limit = 2.2
Sodium, total	4250	mg/kg total solids	EPA 6010B	11/14/14	JHB	Detection limit = 470
Zinc, total	1070	mg/kg total solids	EPA 6010B	11/14/14	JHB	Detection limit = 9.3
Chloride	5630	mg/kg total solids	EPA 300.0A	11/13/14	ALK	Detection limit = 190
Nitrogen, ammonia	32,600	mg/kg total solids	EPA 350.1	11/05/14	JWW	Detection limit = 750
Nitrogen, nitrate	<38	mg/kg total solids	EPA 353.2	11/05/14	MHK	
Nitrogen, total kjeldahl	71,200	mg/kg total solids	EPA 351.2	11/06/14	ALK	Detection limit = 3500
Phosphorus, total (as P)	38,500	mg/kg total solids	SM 4500-P B5,E	11/06/14	JHB	Detection limit = 3800
Sulfate	1470	mg/kg total solids	EPA 300.0A	11/13/14	ALK	Detection limit = 380
Waste pH	7.8	S.U.	EPA 9045C	11/05/14	JWW	
Solids, total	2.70	% by weight	EPA 160.3	11/11/14	EIF	Detection limit = 0.01
Solids, volatile	57.62	% of total solids	EPA 160.4	11/11/14	EIF	Detection limit = 0.01

KAR Laboratories, Inc.

LABORATORY DETAIL REPORT

Client: *City of Cadillac, Water Resources*

KAR Project No. : **404929**

Date Reported : **07/21/14**

Attest: 
David R. Alkema, Lab Manager

Project

Description: *Analysis of one Biosolids sample.*

Sample ID : **"Digester #3"**

Sampled By : *DL of City of Cadillac, Water Resources*

Date Received : **07/11/14**

Sample Date : **07/10/14**

Sample Type : **sludge**

Sample Time : **1320**

KAR Sample No. : **404929-01**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Prep, Hg	Completed		EPA 7471A	07/16/14	BPK	
Prep, metals	Completed		EPA 200.7/8	07/14/14	ZZW	
Aluminum, total	4390	mg/kg total solids	EPA 6010B	07/18/14	JHB	Detection limit = 68
Arsenic, total	<68	mg/kg total solids	EPA 6010B	07/18/14	JHB	
Cadmium, total	<3.4	mg/kg total solids	EPA 6010B	07/18/14	JHB	
Calcium, total	23,300	mg/kg total solids	EPA 6010B	07/18/14	JHB	Detection limit = 68
Chromium, total	78.6	mg/kg total solids	EPA 6010B	07/18/14	JHB	Detection limit = 6.8
Copper, total	736	mg/kg total solids	EPA 6010B	07/18/14	JHB	Detection limit = 14
Lead, total, by ICP	<34	mg/kg total solids	EPA 6010B	07/18/14	JHB	
Magnesium, total	3780	mg/kg total solids	EPA 6010B	07/18/14	JHB	Detection limit = 68
Mercury, total	0.48	mg/kg total solids	EPA 7471A	07/16/14	BPK	Detection limit = 0.15
Molybdenum, total	14.6	mg/kg total solids	EPA 6010B	07/18/14	JHB	Detection limit = 13.6
Nickel, total	13.7	mg/kg total solids	EPA 6010B	07/18/14	JHB	Detection limit = 13.6
Potassium, total	2900	mg/kg total solids	EPA 6010B	07/18/14	JHB	Detection limit = 68
Selenium, total	<68	mg/kg total solids	EPA 6010B	07/18/14	JHB	
Silver, total	<3.6	mg/kg total solids	EPA 6010B	07/15/14	JHB	
Sodium, total	3590	mg/kg total solids	EPA 6010B	07/18/14	JHB	Detection limit = 680
Zinc, total	1200	mg/kg total solids	EPA 6010B	07/18/14	JHB	Detection limit = 14
Chloride	5530	mg/kg total solids	EPA 300.0A	07/16/14	ALK	Detection limit = 150
Nitrogen, ammonia	32,200	mg/kg total solids	EPA 350.1	07/16/14	JWW	Detection limit = 5800
Nitrogen, nitrate	<286	mg/kg total solids	EPA 353.2	07/18/14	ALK	
Nitrogen, total kjeldahl	81,200	mg/kg total solids	EPA 353.2	07/18/14	ALK	Detection limit = 2700
Phosphorus, total (as P)	36,000	mg/kg total solids	SM 4500-P B5,E	07/18/14	JHB	Detection limit = 2900
Sulfate	634	mg/kg total solids	EPA 300.0A	07/16/14	ALK	Detection limit = 290
Waste pH	7.9	S.U.	EPA 9045C	07/16/14	AMG	
Solids, total	3.50	% by weight	EPA 160.3	07/14/14	AMG	Detection limit = 0.01
Solids, volatile	54.60	% of total solids	EPA 160.4	07/14/14	AMG	Detection limit = 0.01

KAR Laboratories, Inc.

LABORATORY DETAIL REPORT

Client: *City of Cadillac, Water Resources*

KAR Project No. : **402028**

Date Reported : **04/15/14**

Attest: 
David R. Alkema, Lab Manager

Project

Description: *Analysis of one Spring Biosolids sample.*

Sample ID : **"Digester #4"**

Sampled By : *DC of City of Cadillac, Water Resources*

Date Received : **04/01/14**

Sample Date : **03/31/14**

Sample Type : **sludge**

Sample Time : **1300**

KAR Sample No. : **402028-01**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Prep, Hg	Completed		EPA 7471A	04/14/14	BPK	
Prep, metals	Completed		EPA 200.7/8	04/03/14	ZZW	
Aluminum, total	2840	mg/kg total solids	EPA 6010B	04/11/14	JHB	Detection limit = 46
Arsenic, total	15.3	mg/kg total solids	EPA 200.8	04/08/14	ZZW	Detection limit = 1
Cadmium, total	2.7	mg/kg total solids	EPA 200.8	04/08/14	ZZW	Detection limit = 0.2
Calcium, total	18,900	mg/kg total solids	EPA 6010B	04/11/14	JHB	Detection limit = 46
Chromium, total	81.8	mg/kg total solids	EPA 6010B	04/11/14	JHB	Detection limit = 4.6
Copper, total	706	mg/kg total solids	EPA 6010B	04/11/14	JHB	Detection limit = 9.1
Lead, total, by ICP	<50	mg/kg total solids	EPA 200.8	04/15/14	ZZW	
Magnesium, total	3040	mg/kg total solids	EPA 6010B	04/11/14	JHB	Detection limit = 46
Mercury, total	0.72	mg/kg total solids	EPA 7471A	04/14/14	BPK	Detection limit = 0.05
Molybdenum, total	23.0	mg/kg total solids	EPA 6010B	04/11/14	JHB	Detection limit = 9.1
Nickel, total	18.6	mg/kg total solids	EPA 6010B	04/11/14	JHB	Detection limit = 9.1
Potassium, total	90.8	mg/kg total solids	EPA 6010B	04/11/14	JHB	Detection limit = 46
Selenium, total	4.2	mg/kg total solids	EPA 200.8	04/08/14	ZZW	Detection limit = 1
Silver, total	<4.6	mg/kg total solids	EPA 6010B	04/14/14	JHB	
Sodium, total	4630	mg/kg total solids	EPA 6010B	04/11/14	JHB	Detection limit = 46
Zinc, total	1060	mg/kg total solids	EPA 6010B	04/11/14	JHB	Detection limit = 9.1
TCLP extraction	Completed		EPA 1311	04/02/14	JWW	
TCLP report	See comment			04/15/14	SRM	This material does not exhibit the Toxicity Characteristic with respect to the requested TC parameters.
ZHE extraction	Completed		EPA 1311	04/02/14	JWW	
Chloride	6780	mg/kg total solids	EPA 300.0A	04/04/14	ALK	Detection limit = 190
Nitrogen, ammonia	36,100	mg/kg total solids	EPA 350.1	04/02/14	JWW	Detection limit = 1500
Nitrogen, nitrate	<37	mg/kg total solids	EPA 353.2	04/03/14	ALK	
Nitrogen, total kjeldahl	81,100	mg/kg total solids	EPA 351.2	04/07/14	ALK	Detection limit = 2600
Phosphorus, total (as P)	41,800	mg/kg total solids	SM 4500-P B5,E	04/08/14	JHB	Detection limit = 3000
Sulfate	2110	mg/kg total solids	EPA 300.0A	04/04/14	ALK	Detection limit = 370
Waste pH	7.7	S.U.	EPA 9045C	04/09/14	AMG	
Solids, total	2.74	% by weight	EPA 160.3	04/03/14	AMG	Detection limit = 0.01
Solids, volatile	57.70	% of total solids	EPA 160.4	04/03/14	AMG	Detection limit = 0.01
Prior. Poll. volatiles	See below		EPA 8260	04/02/14	JAR	
Prep, VOA	Completed		EPA 5030	04/02/14	JAR	
1,1,1-Trichloroethane	<2000	ug/kg total solids	EPA 8260	04/02/14	JAR	
1,1,2,2-Tetrachloroethane	<2000	ug/kg total solids	EPA 8260	04/02/14	JAR	
1,1,2-Trichloroethane	<2000	ug/kg total solids	EPA 8260	04/02/14	JAR	
1,1-Dichloroethane	<2000	ug/kg total solids	EPA 8260	04/02/14	JAR	
1,1-Dichloroethene	<2000	ug/kg total solids	EPA 8260	04/02/14	JAR	
1,2-Dichlorobenzene	<2000	ug/kg total solids	EPA 8260	04/02/14	JAR	
1,2-Dichloroethane	<2000	ug/kg total solids	EPA 8260	04/02/14	JAR	
1,2-Dichloropropane	<2000	ug/kg total solids	EPA 8260	04/02/14	JAR	
1,3-Dichlorobenzene	<2000	ug/kg total solids	EPA 8260	04/02/14	JAR	
1,4-Dichlorobenzene	<2000	ug/kg total solids	EPA 8260	04/02/14	JAR	

KAR Laboratories, Inc.

LABORATORY DETAIL REPORT

Client: **City of Cadillac, Water Resources**

KAR Project No. : **402028**

Date Reported : **04/15/14**

Attest: 
David R. Alkema, Lab Manager

Project

Description: **Analysis of one Spring Biosolids sample.**

Sample ID : **"Digester #4"**

Sampled By : **DC of City of Cadillac, Water Resources**

Date Received : **04/01/14**

Sample Date : **03/31/14**

Sample Type : **sludge**

Sample Time : **1300**

KAR Sample No. : **402028-01**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
2-Chloroethylvinyl ether	<20,000	ug/kg total solids	EPA 8260	04/02/14	JAR	
Acrolein	<10,000	ug/kg total solids	EPA 8260	04/02/14	JAR	
Acrylonitrile	<4000	ug/kg total solids	EPA 8260	04/02/14	JAR	
Benzene	<2000	ug/kg total solids	EPA 8260	04/02/14	JAR	
Bromodichloromethane	<2000	ug/kg total solids	EPA 8260	04/02/14	JAR	
Bromoform	<2000	ug/kg total solids	EPA 8260	04/02/14	JAR	
Bromomethane	<2000	ug/kg total solids	EPA 8260	04/02/14	JAR	
Carbon tetrachloride	<2000	ug/kg total solids	EPA 8260	04/02/14	JAR	
Chlorobenzene	<2000	ug/kg total solids	EPA 8260	04/02/14	JAR	
Chloroethane	<2000	ug/kg total solids	EPA 8260	04/02/14	JAR	
Chloroform	<2000	ug/kg total solids	EPA 8260	04/02/14	JAR	
Chloromethane	<2000	ug/kg total solids	EPA 8260	04/02/14	JAR	
Cis-1,3-Dichloropropene	<2000	ug/kg total solids	EPA 8260	04/02/14	JAR	
Dibromochloromethane	<2000	ug/kg total solids	EPA 8260	04/02/14	JAR	
Ethylbenzene	<2000	ug/kg total solids	EPA 8260	04/02/14	JAR	
Methylene chloride	<2000	ug/kg total solids	EPA 8260	04/02/14	JAR	
Tetrachloroethene	<2000	ug/kg total solids	EPA 8260	04/02/14	JAR	
Toluene	<2000	ug/kg total solids	EPA 8260	04/02/14	JAR	
Trans-1,2-Dichloroethene	<2000	ug/kg total solids	EPA 8260	04/02/14	JAR	
Trans-1,3-Dichloropropene	<2000	ug/kg total solids	EPA 8260	04/02/14	JAR	
Trichloroethene	<2000	ug/kg total solids	EPA 8260	04/02/14	JAR	
Trichlorofluoromethane	<2000	ug/kg total solids	EPA 8260	04/02/14	JAR	
Vinyl chloride	<2000	ug/kg total solids	EPA 8260	04/02/14	JAR	
Prior. Poll. acids	See below		EPA 8270	04/09/14	GMB	
Prior. Poll. base-neutrals	See below		EPA 8270	04/09/14	GMB	
Prep. SV Acid/BN	Completed		EPA 3545	04/08/14	GMB	
1,2,4-Trichlorobenzene 8270	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
1,2-Dichlorobenzene by 8270	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
1,2-Diphenylhydrazine	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
1,3-Dichlorobenzene by 8270	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
1,4-Dichlorobenzene by 8270	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
2,3,7,8-TCDD by 8270	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
2,4,6-Trichlorophenol	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
2,4-Dichlorophenol	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
2,4-Dimethylphenol	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
2,4-Dinitrophenol	<28,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
2,4-Dinitrotoluene	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
2,6-Dinitrotoluene	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
2-Chloronaphthalene	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
2-Chlorophenol	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
2-Methyl-4,6-dinitrophenol	<28,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
2-Nitrophenol	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
3,3'-Dichlorobenzidine	<28,000	ug/kg total solids	EPA 8270	04/09/14	GMB	

KAR Laboratories, Inc.

LABORATORY DETAIL REPORT

Client: **City of Cadillac, Water Resources**

KAR Project No. : **402028**

Date Reported : **04/15/14**

Attest: 
David R. Alkema, Lab Manager

Project

Description: **Analysis of one Spring Biosolids sample.**

Sample ID : **"Digester #4"**

Sampled By : **DC of City of Cadillac, Water Resources**

Date Received : **04/01/14**

Sample Date : **03/31/14**

Sample Type : **sludge**

Sample Time : **1300**

KAR Sample No. : **402028-01**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
4-Bromophenyl phenyl ether	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
4-Chloro-3-methylphenol	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
4-Chlorophenyl phenyl ether	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
4-Nitrophenol	<28,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Acenaphthene	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Acenaphthylene	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Anthracene	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Benzidine	<70,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Benzo(a)anthracene	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Benzo(a)pyrene	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Benzo(b)fluoranthene	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Benzo(ghi)perylene	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Benzo(k)fluoranthene	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Bis(2-chloroethoxy)methane	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Bis(2-chloroethyl)ether	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Bis(2-chloroisopropyl)ether	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Bis(2-ethylhexyl)phthalate	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Butylbenzyl phthalate	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Chrysene	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Di-n-butyl phthalate	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Di-n-octyl phthalate	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Dibenzo(ah)anthracene	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Diethyl phthalate	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Dimethyl phthalate	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Fluoranthene	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Fluorene	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Hexachlorobenzene	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Hexachlorobutadiene	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Hexachlorocyclopentadiene	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Hexachloroethane	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Indeno(123cd)pyrene	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Isophorone	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
N-Nitrosodi-n-propylamine	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
N-Nitrosodimethylamine	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
N-Nitrosodiphenylamine	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Naphthalene by Method 8270	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Nitrobenzene	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Pentachlorophenol	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Phenanthrene	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Phenol	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Pyrene	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Prior. Poll. pest./PCB's	See below		EPA 8270	04/09/14	GMB	
4,4'-DDD	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	

KAR Laboratories, Inc.

LABORATORY DETAIL REPORT

Client: *City of Cadillac, Water Resources*

KAR Project No. : **402028**

Date Reported : **04/15/14**

Attest: 
David R. Alkema, Lab Manager

Project

Description: *Analysis of one Spring Biosolids sample.*

Sample ID : **"Digester #4"**

Sampled By : *DC of City of Cadillac, Water Resources*

Date Received : **04/01/14**

Sample Date : **03/31/14**

Sample Type : **sludge**

Sample Time : **1300**

KAR Sample No. : **402028-01**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
4,4'-DDE	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
4,4'-DDT	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Aldrin	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Alpha-BHC	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Beta-BHC	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Chlordane	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Delta-BHC	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Dieldrin	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Endosulfan I	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Endosulfan II	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Endosulfan sulfate	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Endrin	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Endrin aldehyde	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Gamma-BHC	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Heptachlor	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Heptachlor epoxide	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
PCB Aroclor 1016 by 8270	<140,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
PCB Aroclor 1221 by 8270	<140,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
PCB Aroclor 1232 by 8270	<140,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
PCB Aroclor 1242 by 8270	<140,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
PCB Aroclor 1248 by 8270	<140,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
PCB Aroclor 1254 by 8270	<140,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
PCB Aroclor 1260 by 8270	<140,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Toxaphene	<140,000	ug/kg total solids	EPA 8270	04/09/14	GMB	

Sample ID : **TCLP Leachate of Digester #4**

Sampled By :

Date Received : **04/01/14**

Sample Date :

Sample Type : **TCLP**

Sample Time :

KAR Sample No. : **402028-01T**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
TC Metals	Completed		EPA 200.7	04/10/14	BPK	
Prep. Hg	Completed		EPA 7470A	04/03/14	BPK	
Prep. metals	Completed		EPA 200.7/8	04/03/14	ZZW	
Arsenic, total, by ICP	<0.5	mg/L	EPA 200.7	04/10/14	BPK	TC regulatory limit is 5.0 mg/L.
Barium, total	1.37	mg/L	EPA 200.7	04/10/14	BPK	TC regulatory limit is 100 mg/L.
Cadmium, total	<0.025	mg/L	EPA 200.7	04/10/14	BPK	TC regulatory limit is 1.0 mg/L.
Chromium, total	<0.05	mg/L	EPA 200.7	04/10/14	BPK	TC regulatory limit is 5.0 mg/L.
Copper, total	<0.1	mg/L	EPA 200.7	04/10/14	BPK	
Lead, total, by ICP	<0.25	mg/L	EPA 200.7	04/10/14	BPK	TC regulatory limit is 5.0 mg/L.

KAR Laboratories, Inc.

LABORATORY DETAIL REPORT

Client: *City of Cadillac, Water Resources*

KAR Project No. : **402028**

Date Reported : **04/15/14**

Attest:

David R. Alkema
David R. Alkema, Lab Manager

Project

Description: *Analysis of one Spring Biosolids sample.*

Sample ID : **TCLP Leachate of Digester #4**

Sampled By :

Date Received : **04/01/14**

Sample Date :

Sample Type : **TCLP**

Sample Time :

KAR Sample No. : **402028-01T**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Mercury, total	<0.05	mg/L	EPA 7470A	04/03/14	BPK	TC regulatory limit is 0.2 mg/L.
Selenium, total, by ICP	<0.5	mg/L	EPA 200.7	04/10/14	BPK	TC regulatory limit is 1.0 mg/L.
Silver, total	<0.025	mg/L	EPA 200.7	04/04/14	JHB	TC regulatory limit is 5.0 mg/L.
Zinc, total	1.29	mg/L	EPA 200.7	04/10/14	BPK	
TC Semi-Volatiles	See below		EPA 8270	04/03/14	GMB	
Prep, SV Acid/BN	Completed			04/03/14	GMB	
Prep, herbicides	Completed		EPA 8151	04/07/14	GMB	
2,4,5-Trichlorophenol	<0.1	mg/L	EPA 8270	04/03/14	GMB	TC regulatory limit is 400 mg/L.
2,4,6-Trichlorophenol	<0.1	mg/L	EPA 8270	04/03/14	GMB	TC regulatory limit is 2.0 mg/L.
2,4-Dinitrotoluene	<0.1	mg/L	EPA 8270	04/03/14	GMB	TC regulatory limit is 0.13 mg/L.
Cresols	<0.2	mg/L	EPA 8270	04/03/14	GMB	TC regulatory limit is 200 mg/L.
Hexachlorobenzene	<0.1	mg/L	EPA 8270	04/03/14	GMB	TC regulatory limit is 0.13 mg/L.
Hexachlorobutadiene	<0.1	mg/L	EPA 8270	04/03/14	GMB	TC regulatory limit is 0.5 mg/L.
Hexachloroethane	<0.1	mg/L	EPA 8270	04/03/14	GMB	TC regulatory limit is 3.0 mg/L.
Nitrobenzene	<0.1	mg/L	EPA 8270	04/03/14	GMB	TC regulatory limit is 2.0 mg/L.
Pentachlorophenol	<0.1	mg/L	EPA 8270	04/03/14	GMB	TC regulatory limit is 100 mg/L.
Pyridine	<0.2	mg/L	EPA 8270	04/03/14	GMB	TC regulatory limit is 5.0 mg/L.
TC Pesticides by ECD	See below		EPA 8081	04/04/14	GMB	
Prep, ECD	Completed		EPA 3510	04/03/14	GMB	
Chlordane	<0.02	mg/L	EPA 8081	04/04/14	GMB	TC regulatory limit is 0.03 mg/L.
Endrin	<0.004	mg/L	EPA 8081	04/04/14	GMB	TC regulatory limit is 0.02 mg/L.
Heptachlor	<0.004	mg/L	EPA 8081	04/04/14	GMB	TC regulatory limit is 0.008 mg/L.
Heptachlor epoxide	<0.004	mg/L	EPA 8081	04/04/14	GMB	TC regulatory limit is 0.008 mg/L.
Lindane by ECD	<0.004	mg/L	EPA 8081	04/04/14	GMB	TC regulatory limit is 10.0 mg/L.
Methoxychlor	<0.02	mg/L	EPA 8081	04/04/14	GMB	TC regulatory limit is 0.5 mg/L.
Toxaphene	<0.04	mg/L	EPA 8081	04/04/14	GMB	TC regulatory limit is 0.5 mg/L.
TC Herbicides	See below		EPA 8151	04/08/14	GMB	
2,4,5-TP (Silvex)	<0.2	mg/L	EPA 8151	04/08/14	GMB	TC regulatory limit is 1.0 mg/L.
2,4-D	<0.2	mg/L	EPA 8151	04/08/14	GMB	TC regulatory limit is 10.0 mg/L.

Sample ID : **ZHE Leachate of Digester #4**

Sampled By :

Date Received : **04/01/14**

Sample Date :

Sample Type : **ZHE**

Sample Time :

KAR Sample No. : **402028-01Z**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
TC Volatiles	See below		EPA 8260	04/03/14	JAR	
Prep, VOA	Completed		EPA 5030	04/03/14	JAR	
1,1-Dichloroethene	<0.1	mg/L	EPA 8260	04/03/14	JAR	TC regulatory limit is 0.7 mg/L.
1,2-Dichloroethane	<0.1	mg/L	EPA 8260	04/03/14	JAR	TC regulatory limit is 0.5 mg/L.

KAR Laboratories, Inc.

LABORATORY DETAIL REPORT

Client: *City of Cadillac, Water Resources*

KAR Project No. : **402028**

Date Reported : **04/15/14**

Attest: 
David R. Alkema, Lab Manager

Project

Description: *Analysis of one Spring Biosolids sample.*

Sample ID : **ZHE Leachate of Digester #4**

Sampled By :

Date Received : **04/01/14**

Sample Date :

Sample Type : **ZHE**

Sample Time :

KAR Sample No. : **402028-01Z**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
1,4-Dichlorobenzene	<0.1	mg/L	EPA 8260	04/03/14	JAR	TC regulatory limit is 7.5 mg/L.
Benzene	<0.1	mg/L	EPA 8260	04/03/14	JAR	TC regulatory limit is 0.5 mg/L.
Carbon tetrachloride	<0.1	mg/L	EPA 8260	04/03/14	JAR	TC regulatory limit is 0.5 mg/L.
Chlorobenzene	<0.1	mg/L	EPA 8260	04/03/14	JAR	TC regulatory limit is 100 mg/L.
Chloroform	<0.1	mg/L	EPA 8260	04/03/14	JAR	TC regulatory limit is 6.0 mg/L.
Methyl ethyl ketone	<2	mg/L	EPA 8260	04/03/14	JAR	TC regulatory limit is 200 mg/L.
Tetrachloroethene	<0.1	mg/L	EPA 8260	04/03/14	JAR	TC regulatory limit is 0.7 mg/L.
Trichloroethene	<0.1	mg/L	EPA 8260	04/03/14	JAR	TC regulatory limit is 0.5 mg/L.
Vinyl chloride	<0.1	mg/L	EPA 8260	04/03/14	JAR	TC regulatory limit is 0.2 mg/L.